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APPLICATION EXAMPLE

Determining the active ET 200SP R1 interface module

S7-1500H / GetActiveR1Im / V2.0

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1. Introduction

1.1. Overview

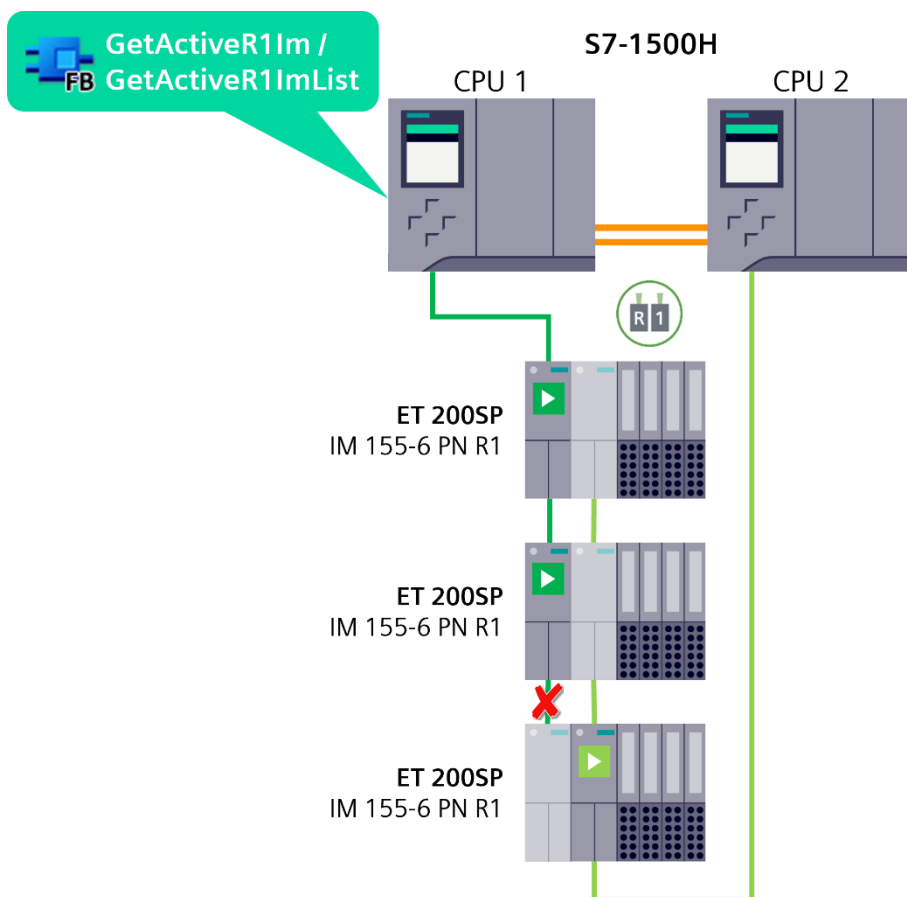
Since the introduction of the interface module IM 155-6 PN R1 for ET 200SP with TIA Portal V18 and from CPU firmware V3.0, you can connect a decentralized IO system to a redundant SIMATIC S7-1500H system via PROFINET system redundancy R1.

This enables highly available control of the decentralized IO system to ensure the function of the automation solution in the following scenarios:

- Failure of an H CPU
- Failure of an interface module
- Interruption of a PROFINET connection (see figure)
- Outage of a subnet

The switchover in the event of a fault takes place automatically.

In order to be informed at all times in the user program about the status of the redundant connection and the active IM of the ET 200SP, the function block **"GetActiveR1Im"** is provided with this application example:



In order to execute the diagnostic block sequentially for a larger number of stations, an envelope block **"GetActiveR1ImList"** is also provided, which can be used to monitor a predefined list of IMs.

1.2. Functionality

The status of the redundant connection of an ET 200SP R1 is determined using the "GetActiveR1Im" function block provided with this application example.

When executing the block, it determines the PROFINET-data record "ARData" (63,520) by means of the internally called system instruction "RDREC". This dataset contains the following relevant information:

- Active Interface Module (IM)
- Status of the redundant connection

The type of topology does not affect the function of the device.

1.3. Components used

This application example was created with these hardware and software components:

Component	Number	Article	Hint
SIMATIC S7-1500 CPU 1517H-3 PN	1	6ES7517-3HP00-0AB0	All SIMATIC S7-1500H/HF possible, from FW 3.0
SIMATIC ET 200SPIM 155-6 PN R1	3	6ES7155-6AU00-0HM0	Including corresponding base units and server modules
TIA Portal STEP 7 Professional V19	1	6ES7822-1AE23-0YA5	Possible from TIA Portal V18

The listed components can be obtained, for example, from the [Siemens Industry Mall](#).

This use case consists of the following components:

Component	Filename	Hint
TIA Portal Project	109963780_GetActiveR1Im_PROJ_V2_0.zip	Example project for TIA Portal V19
SCL function block source	109963780_GetActiveR1Im_CODE_V2_0.zip	SCL sources of the GetActiveR1Im and GetActiveR1ImList devices

HINT

If you are using an older version of TIA Portal, you can generate the function block from the provided SCL sources.

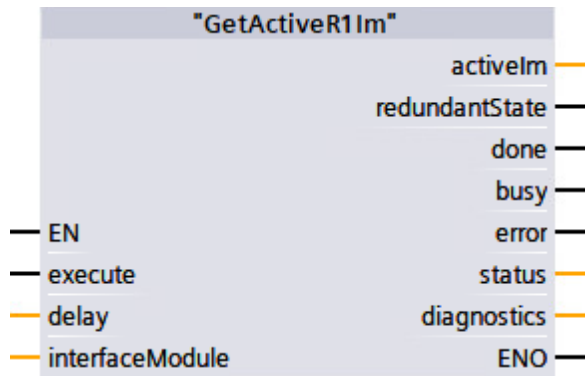
2. Solution

2.1. FB "GetActiveR1Im" for single diagnosis

The function block "GetActiveR1Im" for the one-time diagnosis of a station is described below.
 The processing of the device is asynchronous and can therefore extend over several PLC cycles.

2.1.1. Interface description

The following illustration shows the function block:



The following table describes the function block parameters:

Parameter	Type	Data type	Description
execute	Input	Bool	Execution of the function with a rising flank
delay	Input	Time	(Hidden parameter) Time delay in the execution of the module, see chapter 2.1.3
interfaceModule	Input	HW_INTERFACE	Addressing the IM. Here you can specify any interface of the IM 155-6 PN R1.
activeIm	Output	USInt	Active interface module of the station: "0": IM unknown "1": Left IM "2": Right IM
redundantState	Output	Bool	"TRUE": Redundant connection of the station active
done	Output	Bool	"TRUE": Function executed successfully
busy	Output	Bool	"TRUE": Function in execution
error	Output	Bool	"TRUE": Error on execution
status	Output	Word	Status and error codes, see 2.1.2
diagnostics	Output	"typeDiagnostics"	For extended diagnostic information, see 2.1.2

HINT

The function block can only determine the status for an ET 200SP station. This means that in order to monitor "n" stations, the module must be called or executed "n-times".

Alternatively, the envelope "GetActiveR1ImList" can be used.

2.1.2. Error handling and status

The device uses a standardized diagnostic concept according to the programming style guide for SIMATIC S7-1200/1500.

The "done", "busy", "error" and "status" diagnostic outputs output the current status and errors, while the "diagnostics" output provides a diagnostic structure with detailed information in the event of an error.

HINT

Reset behavior of the outputs for blocks with "execute"

The diagnostic outputs retain their value while the "execute" input is set. If the "execute" input is reset before the processing of the FB is completed, the values of the output parameters are output for one cycle after the job is processed.

Status output interaction

The following table describes the interaction of the status outputs for the FB "GetActiveR1Im":

Done	Busy	error	Description
FALSE	FALSE	FALSE	FB is inactive; waiting for a positive edge for execution
FALSE	TRUE	FALSE	FB performs its function
TRUE	FALSE	FALSE	Function executed without errors; waiting for new flank to execute
FALSE	FALSE	TRUE	An error has occurred that the FB cannot automatically fix, and the FB has aborted the execution of its function (e.g. in the case of invalid input parameters). After the cause has been resolved, a new flank at the "execute" input is necessary.

Status/Error Codes

The following table describes the values of the initial "status"

Code	Description
16#0000	Job completed without error
16#7000	No job active
16#7001	First call of the command
16#7002	Follow-up call of the command
16#7005	Request has been sent, response pending
16#8600	Internal error. PN record cannot be interpreted. If necessary, the output "diagnostics" outputs further status codes of the internal SFB "RDREC".

Output "diagnostics"

In the event of an error, the "diagnostics" output outputs detailed information about the error at hand.

The values are written only when an error occurs and are held until a new job is started or another error has occurred and overwrites the values.

The "diagnostics" output uses the PLC data type "typeDiagnostics", which is structured as follows:

Name	Data type	Description
status	Word	Status of the FB at the time of the error.
subfunctionStatus	DWord	Status or returned value from internal commands or FBs where an error occurred. For detailed information, refer to the online help for the respective command or the documentation of the respective FB.
stateNumber	Dint	State of the internal state machine in which the error occurred.

2.1.3. Call Environment

Due to the acyclic processing of the function block, we recommend calling it in the cyclic user program (e.g. Main/OB1) of the control system.

Call in case of error

If the device is to be used for evaluation in the event of an error, it is recommended to set the input parameter "execute" in a suitable error OB:

- "Hardware interrupt" (OB40)
- "IO redundancy error" (OB70)*
- "CPU redundancy error" (OB72)
- "Diagnostic error interrupt" (OB82)
- "IO access error" (OB122):

* Recommendation: In order to start the processing of the module only with redundantly connected peripherals, the use of OB70 is recommended. The other error OBs listed here are also executed due to other causes of errors.

Delay

Since the generation and reading of the PROFINET data record takes some time after a connection interruption, we recommend that the device be executed with a time delay in order to avoid inconsistent reading results.

HINT

We recommend a time delay ("delay") of at least 50 ms ("T#50ms") to ensure that the PROFINET dataset is read consistently. The time delay may need to be increased depending on the components used, the configuration, and the network.

Repeated execution

In order to continuously monitor the status of your redundant decentralized peripherals, the block can be executed on a scheduled basis (e.g. by timer). Pay attention to the diagnostic outputs of the block in order to start the execution again only after the defined completion ("done" = "TRUE" or "error" = "TRUE") of the previous execution.

In this case, the use of the function block "GetActiveR1ImList" is preferred anyway.

2.2. FB "GetActiveR1ImList" for sequential diagnostics

To continuously monitor one or more R1 stations, the function block "GetActiveR1ImList" can be used.

2.2.1. Functionality

The underlying single block "GetActiveR1Im" is called internally as a multi-instance – but with the following extensions:

- The envelope allows you to specify a list of interface modules, not just a single module.
- All valid entries in the list are processed sequentially, i.e. the status of the corresponding interface modules is queried one after the other.
- The envelope block enables the internal instantiation of several multi-instances of the single module "GetActiveR1Im" working in parallel, which contribute to a faster processing of the specified list of interface modules.
- In order to carry out the monitoring continuously, the envelope has an "enable" behavior according to the PLCopen standard. The list of IMs is cyclically traversed as long as the "Enable" input is set.
- A call with "Execute" behavior, i.e. the one-time check of all listed stations in the event of an error, is possible. Then the "Delay" parameter has to be set (see [Call Environment](#)). However, it is recommended to choose continuous monitoring ("enable") in order to reliably detect any missed state changes due to repeated execution.
- An additional output indicates whether all specified stations are in a redundant state. This makes it possible to see at a glance whether the planned R1 system redundancy is fully available across all stations.

2.2.2. Interface description

The "GetActiveR1ImList" module works according to an "enable" behavior and is based on the template of the programming style guide for S7-1200 and S7-1500.

The following illustration shows the function block:



Parameter	Parameter	Data type	Description
enable	Input	Bool	Permanent activation of the device
execute	Input	Bool	(hidden parameter) one-time execution of the device, see 2.1.3
delay	Input	Time	(Hidden parameter) Time delay in the first execution of the device, see chapter 2.1.3 (only relevant when executed using "execute")
numberOfInstances	Input	USInt	Number of actively used instances of GetActiveR1Im, see section "Scaling"
overallR1RedState	Output	Bool	Collected redundancy state of all specified IMs
valid	Output	Bool	"TRUE": Output of the function block valid
busy	Output	Bool	"TRUE": Function in execution
error	Output	Bool	"TRUE": Error on execution
status	Output	Word	Status and error codes
Diagnostics	Output	"typeDiagnostics"	Advanced diagnostic information
r1Stations	InOut	Array[*] of HW_INTERFACE	List of IMs of the stations to be tested, matching the "interfaceModule" of the individual module "GetActiveR1Im"
r1StationRedStates	InOut	Array[*] of "typeR1StationRedState"	List of the redundancy states of the stations to be tested, analogous to the outputs "activeIm" and "redundantState" of the individual module "GetActiveR1Im"

2.2.3. Status

Status/Error Codes

The following table describes the values of the initial "status"

Code	Description
16#0000	Job completed without error
16#7000	No job active
16#7001	First call of the command
16#7002	Follow-up call of the command
16#8201	Array definition of "r1Stations" and "r1StationRedStates" inconsistent
16#8202	No interface module specified in array for "r1Stations"

2.2.4. Scaling

Number of stations

The stations to be monitored are connected to the module as InOut parameters, i.e. passed as a reference. The array "r1Stations" of the data type "HW_INTERFACE" is defined as an "Array[*]" of unspecified size. The array connected to it can therefore take on any size that matches the number of real stations in the application. Empty entries are possible and are skipped at runtime.

The array "r1StationRedStates" for the returned status information of the stations is defined in the same way. Its boundaries (and thus its size) must match exactly those of the array "r1Stations" of the stations. The sorting of the stations is the same as in the array "r1Stations".

Number of instances

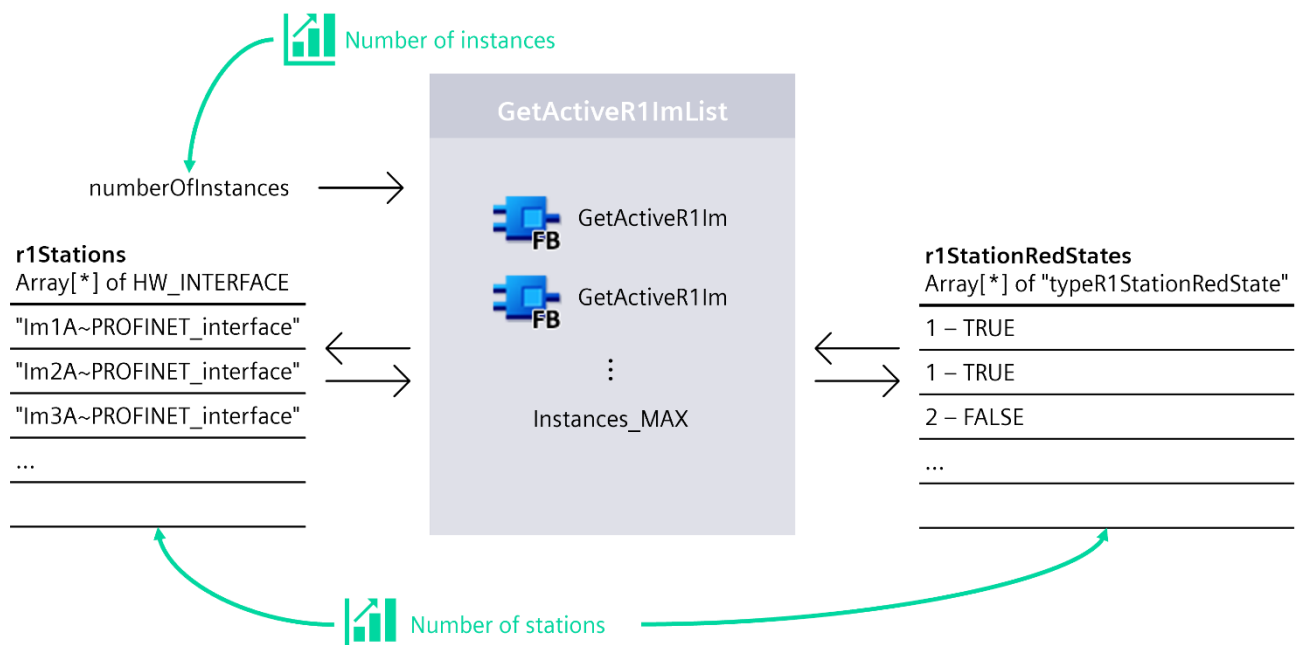
The FB "GetActiveR1ImList" instantiates an array of multi-instances of the "GetActiveR1Im" device. The number of instances can be defined during configuration with the internal constant "INSTANCES_MAX". This value corresponds to the maximum number of instances that can be actively used in the company.

The input parameter "numberOfInstances" can be used at runtime to set how many of these instances are actually to be actively used. The decisive factor here is the value at the time of activation of the module via the input "enable".

ATTENTION

The FB "GetActiveR1Im" uses the mechanism of the PROFINET Data Records internally to obtain the status of the interface modules.

The total number of active instructions "RD_REC" and "WR_REC" is limited to 20 on an S7-1500. This amount must be shared with other tasks of the user program, e.g. for serial communication. It may therefore be advisable to limit the instances assigned to processing at runtime so as not to interfere with higher-priority tasks.



Functioning

The active multi-instances work in parallel. Whenever an instance has finished processing and is ready for a new execution, it is assigned the index of the sequentially next IM to be checked. Once the last valid entry in the list of stations has been processed, "enable" operation will resume without interruption at the beginning of the list.

The repetition rate results solely from the ratio of the number of stations and the number of active instances as well as the call environment of the envelope "GetActiveR1ImList" (and thus the duration of the internal calls of the instances of "GetActiveR1Im"). The parameter "valid" of each output data record of a station indicates in which cycle a new status information is available.

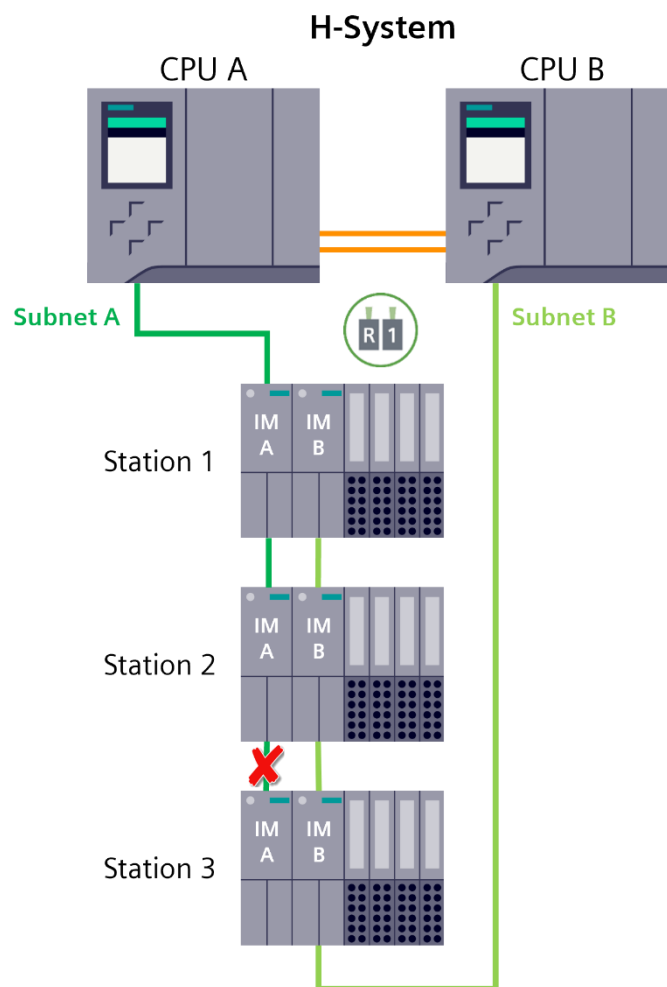
3. Example project

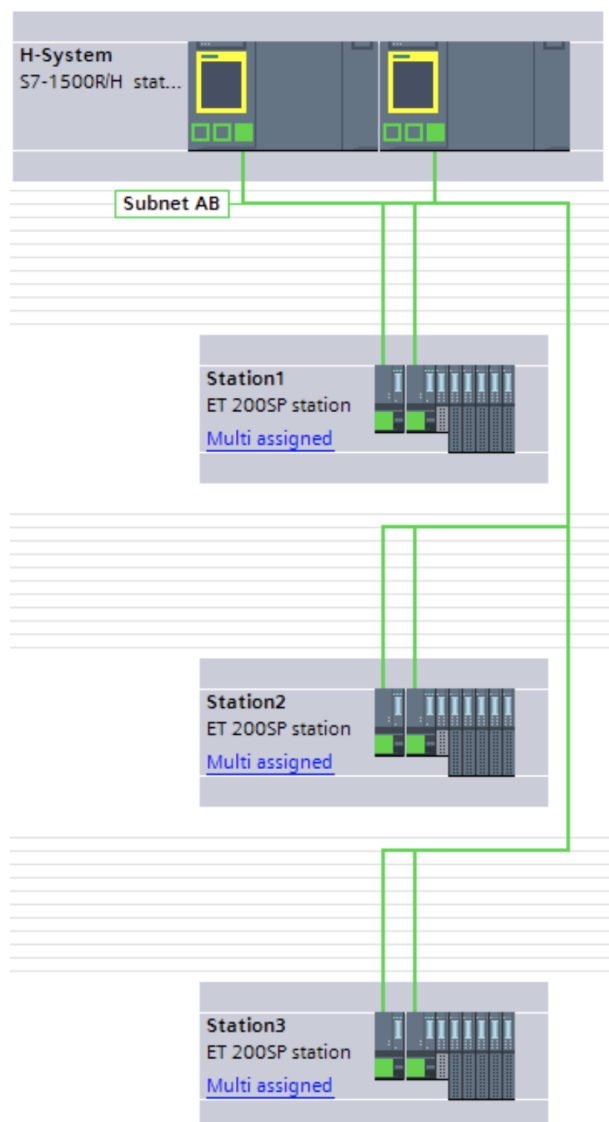
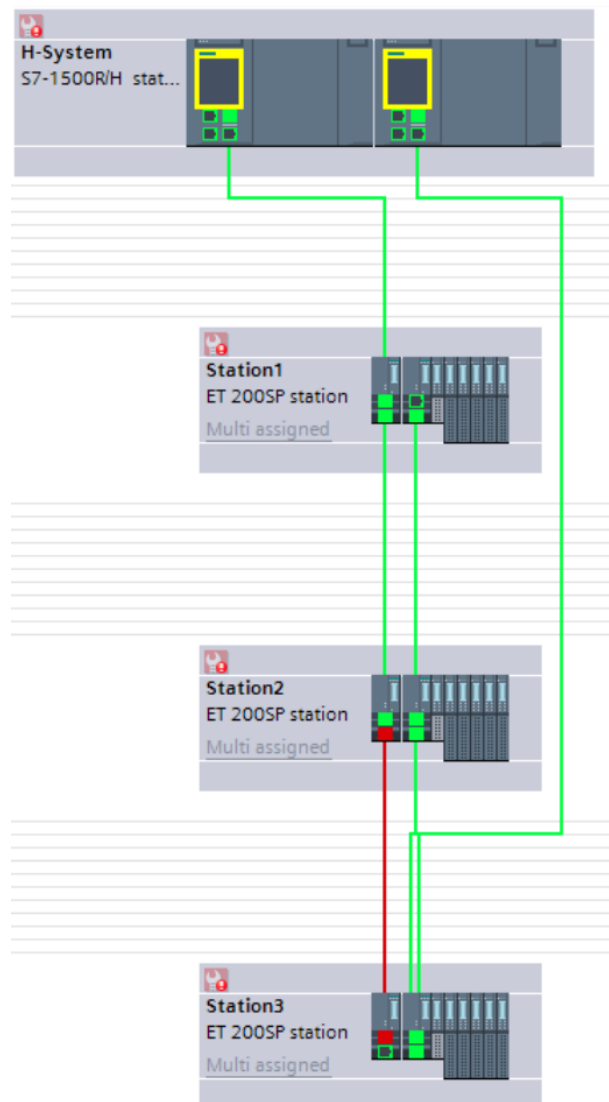
The included sample project is fully configured and programmed to show the functions, implementation and use of the function block "GetActiveR1Im" as an example.

3.1. Hardware configuration

SIMATIC S7-1500H/HF and ET 200SP R1 can be built in line, star or ring topology (with and without MRP rings). To explain the functions of the "GetActiveR1Im" module, the line topology is selected in this application example. The type of topology does not affect the function of the device.

The following figure shows the hardware structure of this application example:



Network view**Topology (optional)**

It is important to ensure that the stations are assigned to the H-System by multiple assignment ("Assign to new DP master/IO controller" in the context menu).

In this application example, all interface modules are located in a common subnet. This is particularly necessary if the R1-based infrastructure is to be expanded with a so-called "Y-switch" (SCALANCE XF204-2BA DNA) to include S2 and S1 devices. Otherwise, it would also be conceivable to set up two separate subnets. Then even identical IP addresses could be assigned to both interface modules of a station.

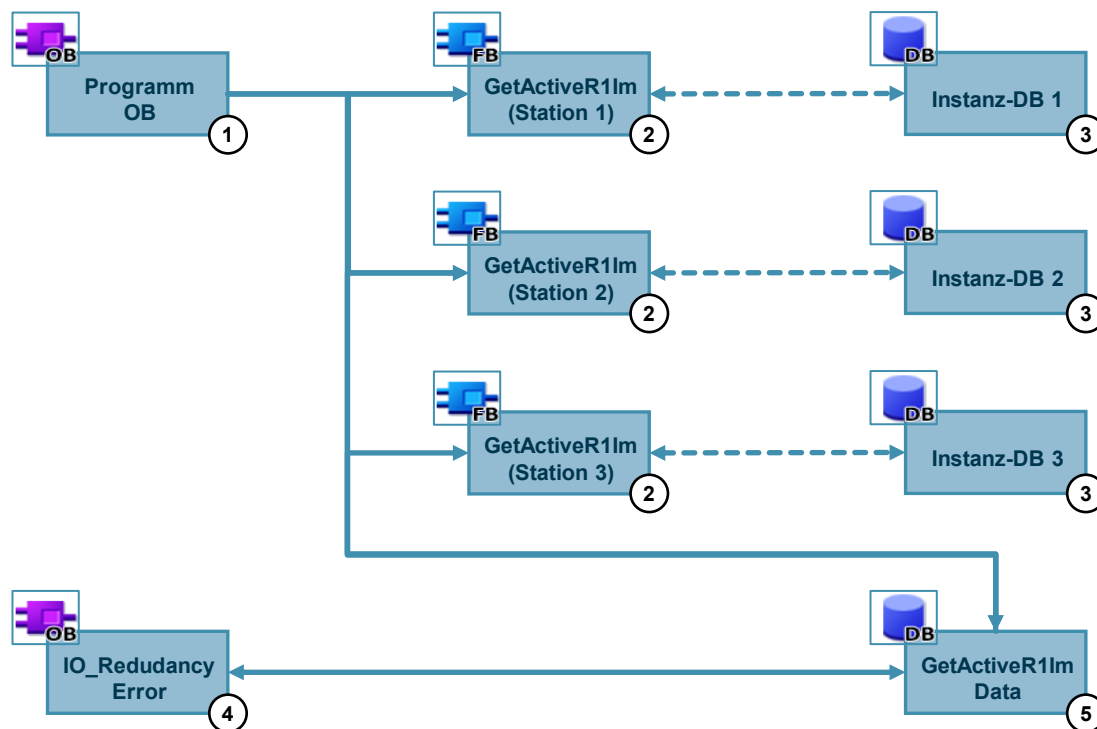
The creation of a topology, i.e. the configuration of the physical connection paths, is optional. However, it can be particularly helpful in the scenario of a device exchange by enabling an automatic naming of the device name of a PROFINET device.

The online view of the topology shows the scenario on which the operation of the modules is to be shown: The connection between the left interface modules of stations 2 and 3 is interrupted. Station 3 must be controlled via the right IM.

3.2. Single diagnosis using "GetActiveR1Im"

3.2.1. User program

The call environment of the individual modules in the program example is shown below:



No.	Description
1.	Program cycle (e.g. OB1 or OB123): Cyclic call to the function block "GetActiveR1ImList"
2.	Function block "GetActiveR1Im", once per ET 200SP R1 station
3.	Instance data block, each per function block "GetActiveR1Im"
4.	IO_RedundancyError (OB70): Automatically sets the trigger variable "execute" for the function blocks when a change is made in the redundant IO system
5.	Data module that contains the input and output parameters for controlling the function blocks

The example project is designed in such a way that the function block "GetActiveR1Im" is executed automatically as soon as a change to the redundant IO system is detected. This is implemented by setting the parameter for executing the block ("execute") in the error OB "IO_RedundancyError" to "TRUE" and thus the processing is started.

All parameters for "GetActiveR1Im" are stored in the data block "DataGetActiveR1Im". For each ET 200SP station in the project, a separate structure is created for the parameters of the function blocks.

3.2.2. Operation

Start execution

The input parameter "execute" can be manually set to "TRUE" to trigger a one-time execution.

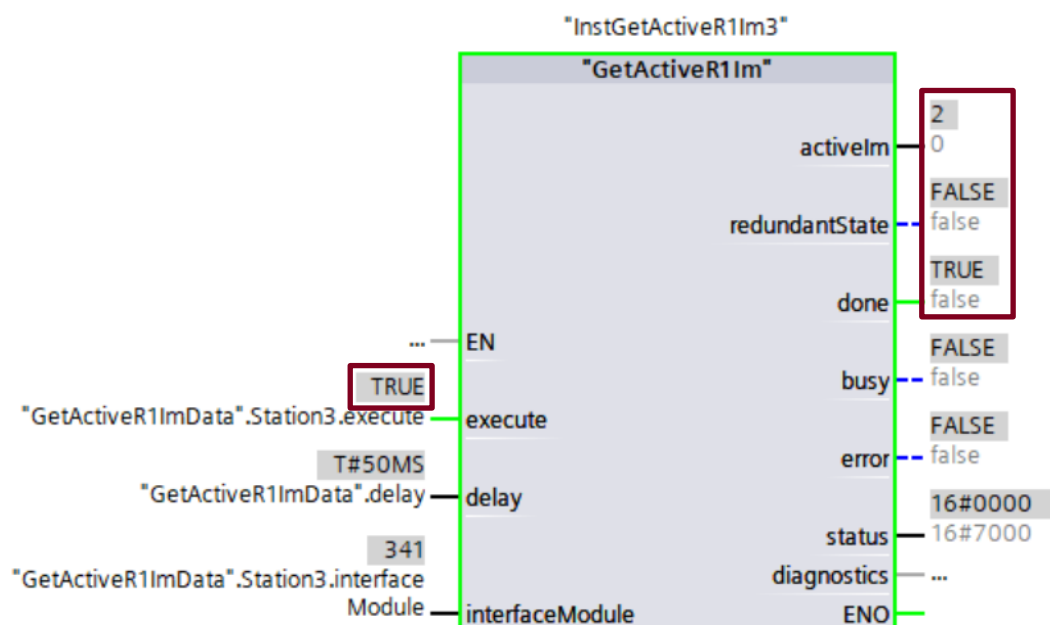
As long as the system is in "RUN-Redundant", "execute" is automatically set by the error OB70 "IO_RedundancyError". This is done, for example, in the following cases:

- Pulling or plugging a cable
- Switching an interface module on or off

Result

The output parameter "done" indicates that the execution was successful and valid values are available.

The right interface module "B" is active ("activelm" = 2). Because the left interface module is not reachable, the "redundantState" is output as "FALSE".



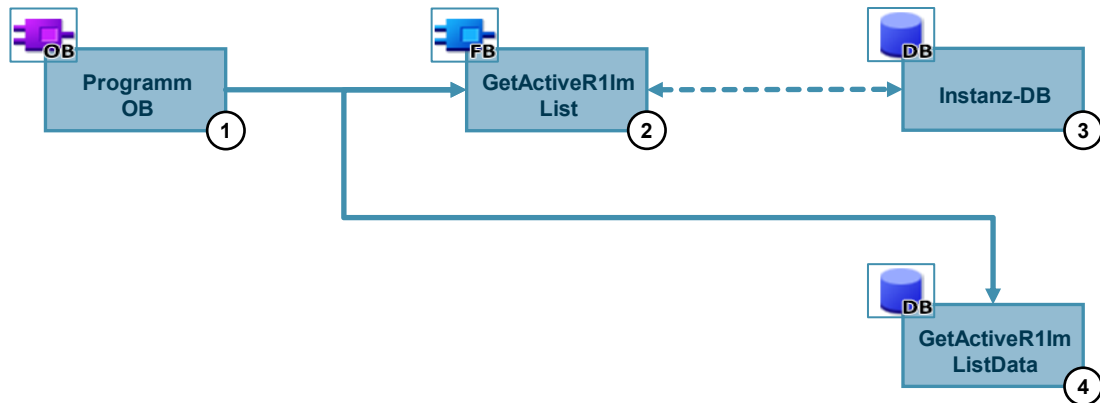
The status of station 3 is also entered in the global data module "GetActiveR1ImData":

GetActiveR1ImData				
	Name	Data type	Start value	Monitor value
Static				
	delay	Time	T#50ms	T#50MS
	Station1	Struct		
	Station2	Struct		
	Station3	Struct		
	execute	Bool	false	FALSE
	interfaceModule	HW_INTERFACE	"Im3A~PROFINET_interface"	341
	r1StationRedState	"typeR1StationRedState"		
	activelm	USInt	0	2
	redundantState	Bool	false	FALSE
	valid	Bool	false	TRUE

3.3. Sequential diagnostics using GetActiveR1ImList

3.3.1. User program

The call environment of the individual blocks in the program example is shown below:



No.	Description
1.	Program cycle (e.g. OB1 or OB124): Cyclic call to the function block "GetActiveR1ImList"
2.	"GetActiveR1ImList" function block, unique for all ET 200SP R1 stations
3.	Instance data block for function block "GetActiveR1ImList"
4.	Data block that contains the input and output parameters for controlling the function block

3.3.2. Operation

Hardware IDs as a list

In the array "r1Stations" in the global data block "GetActiveR1ImListData", the HW_INTERFACE of each interface module of the station to be diagnosed must be entered:

GetActiveR1ImListData			
Name	Data type	Start value	
Static			
enable	Bool	TRUE	
execute	Bool	false	
numberOfInstances	USInt	1	
r1Stations	Array[1..5] of HW_INTERFACE		
r1Stations[1]	HW_INTERFACE	"Im1A~PROFINET_interface"	
r1Stations[2]	HW_INTERFACE	"Im2A~PROFINET_interface"	
r1Stations[3]	HW_INTERFACE	"Im3A~PROFINET_interface"	
r1Stations[4]	HW_INTERFACE	0	
r1Stations[5]	HW_INTERFACE	0	

In addition, the parameter "numberOfInstances" must be used to specify how many instances of "GetActiveR1Im" are to be actively used (see chapter ["Scaling"](#)). In this example, only one instance is used.

The Enable parameter is preset to "TRUE", so that the station check is always executed.

Result

The r1StationRedStates array outputs the status information of the ET 200SP R1 stations.

The "overallR1RedState" indicates with "FALSE" that there is a redundancy failure at at least one of the three stations to be tested.

GetActiveR1ImListData				
Name	Data type	Start value	Monitor value	
Static				
enable	Bool	TRUE	TRUE	
execute	Bool	false	TRUE	
numberOfInstances	USInt	1	1	
r1Stations	Array[1..5] of HW_INTERFACE			
r1StationRedStates	Array[1..5] of "typeR1StationRedState"			
r1StationRedStates[1]	"typeR1StationRedState"			
activeIm	USInt	0	1	
redundantState	Bool	false	TRUE	
valid	Bool	false	FALSE	
r1StationRedStates[2]	"typeR1StationRedState"			
activeIm	USInt	0	1	
redundantState	Bool	false	TRUE	
valid	Bool	false	TRUE	
r1StationRedStates[3]	"typeR1StationRedState"			
activeIm	USInt	0	2	
redundantState	Bool	false	FALSE	
valid	Bool	false	TRUE	
r1StationRedStates[4]	"typeR1StationRedState"			
r1StationRedStates[5]	"typeR1StationRedState"			
overallR1RedState	Bool	false	FALSE	

The left interface module A of station 3 is not available and the active connection to the CPU runs via the right interface module instead. At stations 1 and 2, both interface modules are accessible.

The "valid" fields indicate when a new state value is available (in the cycle at the time of the screenshot shown for stations 2 and 3).

4. Appendix

4.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

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In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- **Support**
In Support, you can find all information helpful for resolving technical issues with our products.
- **mySieportal**
mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

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Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

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For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:



4.2. Links and literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to the contribution page of the application example https://support.industry.siemens.com/cs/ww/en/view/109963780
\3\	SIMATIC S7-1500 Redundant System S7-1500R/H – System Manual https://support.industry.siemens.com/cs/ww/en/view/109754833
\4\	SIMATIC ET 200SP Decentralized Peripheral System – System Manual https://support.industry.siemens.com/cs/ww/en/view/58649293
\5\	Diagnosing a S7-1500R/H redundant system using a function block (R_H_Sys_Status) https://support.industry.siemens.com/cs/ww/en/view/109763768

4.3. Change documentation

Version	Date	Modification
V1.0	07/2024	First edition
V2.0	12/2025	Extension to FB "GetActiveR1ImList", minor improvements to FB "GetActiveR1Im"